

Chemistry · Medium paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ Date _____

117 marks

reading 15 min · writing 150 min

State average on these questions

60 / 117 (51%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. $v = \sqrt{2gh} = 9.9 \text{ m/s}$.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the middle band of questions.

Sources

- [2022 Chemistry exam \(PDF\)](#) · [examiners' report](#) (previous study design)
- [2023 Chemistry exam \(PDF\)](#) · [examiners' report](#) (previous study design)
- [2024 Chemistry exam \(PDF\)](#) · [examiners' report](#)
- [2025 Chemistry exam \(PDF\)](#) · [examiners' report](#)

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–D. One mark each.

Question 1 · 1 mark

2022 VCAA exam · Section A · Question 4 · [previous study design](#) · [open in the official exam PDF \(p. 3\)](#) ↗

Choosing the correctly labelled fuel-cell diagram

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 2 · 1 mark

2022 VCAA exam · Section A · Question 6 · [previous study design](#) · [open in the official exam PDF \(p. 3\)](#) ↗

Comparing energy transformations in two cell types

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 3 · 1 mark

2022 VCAA exam · Section A · Question 14 · [previous study design](#) · [open in the official exam PDF \(p. 6\)](#) ↗

Identifying the reducing agent during recharge of a flow battery

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 4 · 1 mark

2022 VCAA exam · Section A · Question 15 · [previous study design](#) · [open in the official exam PDF \(p. 6\)](#) ↗

Energy released per mole of O₂ in the respiration of glucose

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 5 · 1 mark

2022 VCAA exam · Section A · Question 24 · [previous study design](#) · [open in the official exam PDF \(p. 9\)](#) ↗

Elution order of amino acids with a polar mobile phase

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 6 · 1 mark

2022 VCAA exam · Section A · Question 26 · [previous study design](#) · [open in the official exam PDF \(p. 10\)](#) ↗

Comparing the calibration factors of two calorimeters

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 7 · 1 mark

2022 VCAA exam · Section A · Question 29 · [previous study design](#) · [open in the official exam PDF \(p. 12\)](#) ↗

Ordering carbon–halogen bond strengths

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 8 · 1 mark

2023 VCAA exam · Section A · Question 8 · [previous study design](#) · [open in the official exam PDF \(p. 4\)](#) ↗

K for a doubled equation

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 9 · 1 mark

2023 VCAA exam · Section A · Question 9 · [previous study design](#) · [open in the official exam PDF \(p. 4\)](#) ↗

Evaluating three statements about biofuels

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 10 · 1 mark2023 VCAA exam · Section A · Question 17 · [previous study design](#) · [open in the official exam PDF \(p. 7\)](#) ↗

Mass of fuel needed given energy content and efficiency

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 11 · 1 mark2023 VCAA exam · Section A · Question 26 · [previous study design](#) · [open in the official exam PDF \(p. 10\)](#) ↗

Counting the independent variables in a described experiment

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 12 · 1 mark2024 VCAA exam · Section A · Question 1 · [open in the official exam PDF \(p. 2\)](#) ↗

Classifying photosynthesis (energy and electron transfer)

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 13 · 1 mark2024 VCAA exam · Section A · Question 4 · [open in the official exam PDF \(p. 3\)](#) ↗

Oxidising agent and a conjugate redox pair

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 14 · 1 mark2024 VCAA exam · Section A · Question 5 · [open in the official exam PDF \(p. 3\)](#) ↗

Electron flow and energy change in a galvanic cell with a Pt electrode

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 15 · 1 mark2024 VCAA exam · Section A · Question 8 · [open in the official exam PDF \(p. 4\)](#) ↗

Evaluating two statements about fuel cells

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 16 · 1 mark2024 VCAA exam · Section A · Question 14 · [open in the official exam PDF \(p. 6\)](#) ↗

Conditions giving the highest equilibrium yield for an endothermic reaction

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 17 · 1 mark2024 VCAA exam · Section A · Question 16 · [open in the official exam PDF \(p. 7\)](#) ↗

Comparing forward and reverse rates from a concentration–time graph

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 18 · 1 mark2024 VCAA exam · Section A · Question 20 · [open in the official exam PDF \(p. 8\)](#) ↗

Counting aldehyde isomers of a formula with a single methyl branch

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 19 · 1 mark2024 VCAA exam · Section A · Question 22 · [open in the official exam PDF \(p. 9\)](#) ↗

Molecular formula of a triglyceride formed by condensation

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 20 · 1 mark

2024 VCAA exam · Section A · Question 23 · [open in the official exam PDF \(p. 9\)](#) ↗

Why water is not added during a particular synthesis step

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 21 · 1 mark

2024 VCAA exam · Section A · Question 25 · [open in the official exam PDF \(p. 10\)](#) ↗

Retention order of three C6 compounds

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 22 · 1 mark

2024 VCAA exam · Section A · Question 29 · [open in the official exam PDF \(p. 11\)](#) ↗

Bond type and structural level of disulfide bridges

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 23 · 1 mark

2025 VCAA exam · Section A · Question 6 · [open in the official exam PDF \(p. 3\)](#) ↗

Energy absorbed per mole of CO₂ in photosynthesis

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 24 · 1 mark

2025 VCAA exam · Section A · Question 9 · [open in the official exam PDF \(p. 5\)](#) ↗

Why zinc can be used with an aqueous electrolyte in a cell

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 25 · 1 mark

2025 VCAA exam · Section A · Question 13 · [open in the official exam PDF \(p. 7\)](#) ↗

Half-equation for the reduction during recharge of a rechargeable cell

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 26 · 1 mark

2025 VCAA exam · Section A · Question 15 · [open in the official exam PDF \(p. 8\)](#) ↗

K for a reversed and doubled equation

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 27 · 1 mark

2025 VCAA exam · Section A · Question 17 · [open in the official exam PDF \(p. 9\)](#) ↗

Evaluating two claimed benefits of a fuel-cell design

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 28 · 1 mark

2025 VCAA exam · Section A · Question 25 · [open in the official exam PDF \(p. 13\)](#) ↗

Separating two liquids with similar boiling points

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 29 · 1 mark

2025 VCAA exam · Section A · Question 26 · [open in the official exam PDF \(p. 13\)](#) ↗

Molecular formulas of two terpenes from skeletal structures

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 30 · 1 mark2025 VCAA exam · Section A · Question 29 · [open in the official exam PDF \(p. 14\)](#) ↗Using moles of I_2 to find the number of $C=C$ bonds in a fatYour answer: ☐ A ☐ B ☐ C ☐ D

Section B – Short- and extended-answer questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (7 marks)2022 VCAA exam · Section B · Question 1 · [previous study design](#) · [open in the official exam PDF \(p. 14\)](#) ↗

Pathway from butene to a primary alcohol, then esterification with methanoic acid

a. Type of reaction converting C_4H_8 to the chloroalkane (1 mark)**b.** Semi-structural formula of the isomer that cannot give a primary alcohol (1 mark)**c.** Reagent to convert the chloroalkane into a primary alcohol (1 mark)**d.i.** Catalyst for esterification (1 mark)**d.ii.** Balanced equation for propan-1-ol + methanoic acid (2 marks)**d.iii.** IUPAC name of the ester (1 mark)

Question 2 (10 marks)2022 VCAA exam · Section B · Question 2 · [previous study design](#) · [open in the official exam PDF \(p. 16\)](#) ↗

Carbon capture tank pressure; an acidic electrolyser (electron flow, membrane, Faraday calculation)

a. Pressure in the CO₂ storage tank (2 marks)

b.i. Direction of electron flow in the wire, justified (2 marks)

b.ii. Two functions of the membrane (2 marks)

c.i. Overall equation for the electrolyser at 80 °C (1 mark)

c.ii. Moles of H₂ from 1625.0 A for 1.25 h (3 marks)

Question 3 (10 marks)

2023 VCAA exam · Section B · Question 7 · [previous study design](#) · [open in the official exam PDF \(p. 26\)](#) ↗

Identifying a C/H/O ketone from mass, ¹H NMR, ¹³C NMR and IR spectra

a.i. Molecular formula of the molecule, justified from the mass spectrum (2 marks)

a.ii. Why there is a small peak at m/z = 87 (1 mark)

b. What the doublet at 1.1 ppm tells you (1 mark)

c. Draw a structure consistent with all the data (3 marks)

d. Why one molecule absorbs several IR frequencies (3 marks)

Question 4 (8 marks)

2024 VCAA exam · Section B · Question 1 · [open in the official exam PDF \(p. 13\)](#) ↗

Melting point for purity; linalool: chiral centres, % by mass from I₂ uptake, extraction

a. How melting point determination checks the identity and purity of solids (2 marks)

b.i. Number of chiral centres in linalool (1 mark)

b.ii. % by mass of linalool in a mixture with hexane, from moles of I₂ reacted (3 marks)

b.iii. Two processes to obtain purified linalool from lavender (2 marks)

Question 5 (8 marks)

2024 VCAA exam · Section B · Question 4 · [open in the official exam PDF \(p. 18\)](#) ↗

Haber process equilibrium using partial pressures (K_p), Q vs K , catalyst and green chemistry

a. Expression for K_p (1 mark)

b. Partial pressure of NH_3 at equilibrium (1 mark)

c. A change that makes $Q = 1.20 \times 10^{-9}$ and how the system responds (3 marks)

d. How the iron catalyst resolves the rate–temperature conflict, with a green principle (3 marks)

Question 6 (11 marks)

2024 VCAA exam · Section B · Question 5 · [open in the official exam PDF \(p. 20\)](#) ↗

Heating water with a butane burner (efficiency), then solution calorimetry of a precipitation reaction

a.i. Energy released by burning 0.580 g of butane (2 marks)

a.ii. Energy absorbed by the 100.0 g of water (1 mark)

a.iii. Percentage energy efficiency (2 marks)

b.i. Why a calorimeter must be calibrated (2 marks)

b.ii. Maximum temperature if 0.048 mol of KI reacts (3 marks)

b.iii. A reason the average maximum temperature was higher than expected (1 mark)

Question 7 (13 marks)

2024 VCAA exam · Section B · Question 6 · [open in the official exam PDF \(p. 22\)](#) ↗

Identifying a $\text{C}_5\text{H}_{11}\text{ClO}$ compound from IR, mass, ^{13}C and ^1H NMR spectra

a. Bond responsible for the highest-wavenumber IR absorption (1 mark)

b.i. m/z of the base peak (1 mark)

b.ii. Formula of the species at $m/z = 73$ (1 mark)

c. What four ^{13}C peaks say about five carbons (3 marks)

d.i. What an integration curve represents (1 mark)

d.ii. What the relative sizes of the integration curves tell you (1 mark)

d.iii. What the given integration values indicate (1 mark)

d.iv. What the 0.96 ppm peak (shift and size) indicates (2 marks)

e. Draw Compound X (2 marks)

Question 8 (12 marks)

2024 VCAA exam · Section B · Question 7 · [open in the official exam PDF \(p. 28\)](#) ↗

Electrolysis with Fe and Pt electrodes; recharging a lithium-ion cell; artificial photosynthesis

a.i. Which electrode is the anode (1 mark)

a.ii. Half-equation at the Pt electrode (1 mark)

a.iii. Why no reaction occurs once the power is off (1 mark)

b.i. Polarity of electrode X during recharge (1 mark)

b.ii. Mass of LiC_6 produced at 1.25 A for 1.00 h (4 marks)

c.i. Overall energy transformation in artificial photosynthesis (1 mark)

c.ii. Direction of electron flow on the diagram (1 mark)

c.iii. How charge stays balanced when electrons flow one way (2 marks)

Question 9 (8 marks)

2025 VCAA exam · Section B · Question 7 · [open in the official exam PDF \(p. 36\)](#) ↗

Steelmaking: molten oxide electrolysis vs the blast furnace, judged on sustainability and green chemistry

a. A requirement for molten oxide electrolysis to help UN SDG 12 (2 marks)

b. Evaluate both processes against two green principles, include an ethical factor, and conclude (6 marks)

END OF PAPER

Marking guide follows

Stop here until you have finished the paper under exam conditions.

Marking guide

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	A	Check three things: there is an external load, the fuel is supplied at the anode, and excess fuel leaves from the anode side. Diagram A has all three. <i>Most popular wrong answer: C (24%).</i>	56% correct A 56% · B 7% · C 24% · D 13%
☐	2	B	Both turn chemical energy into electrical energy, and both lose some energy as heat. <i>Most popular wrong answer: D (21%).</i>	56% correct A 8% · B 56% · C 15% · D 21%
☐	3	C	On recharge VO^{2+} is oxidised, and the species that is oxidised is the reducing agent. <i>Most popular wrong answer: D (23%).</i>	53% correct A 10% · B 14% · C 53% · D 23%
☐	4	D	Six O_2 are consumed per glucose: $2805 \div 6 = 467.5 \text{ kJ mol}^{-1}$. <i>Most popular wrong answer: C (22%).</i>	55% correct A 17% · B 5% · C 22% · D 55%
☐	5	D	With a polar mobile phase the most polar compound moves fastest and elutes first: Asn, then Ala, then Leu. <i>Most popular wrong answer: C (22%).</i>	50% correct A 18% · B 10% · C 22% · D 50%
☐	6	C	The same calibration energy was supplied to both. $CF = E/\Delta T$, so the calorimeter with the smaller temperature rise has the larger CF, and that is calorimeter 2. <i>Most popular wrong answer: A (27%).</i>	57% correct A 27% · B 10% · C 57% · D 5%
☐	7	A	Bonds get weaker as the halogen gets bigger (longer bond): $\text{C-Br} < \text{C-Cl} < \text{C-F}$. <i>Most popular wrong answer: D (19%).</i>	54% correct A 54% · B 11% · C 15% · D 19%
☐	8	D	Doubling every coefficient squares K, and the units square too: $K^2 = 16.8 \text{ M}^{-4}$. <i>Most popular wrong answer: C (18%).</i>	52% correct A 15% · B 15% · C 18% · D 52%
☐	9	D	Test each statement against the data. All three hold. <i>Most popular wrong answer: C (38%).</i>	54% correct A 5% · B 3% · C 38% · D 54%
☐	10	D	Divide the useful energy needed by the efficiency, then by the energy content: $42 \div 0.39 \div 19.0 \approx 5.67 \text{ t}$. <i>Most popular wrong answer: C (24%).</i>	57% correct A 9% · B 10% · C 24% · D 57%
☐	11	B	List what the experimenter deliberately changed. There are three. <i>Most popular wrong answer: A (21%).</i>	50% correct A 21% · B 50% · C 20% · D 9%
☐	12	B	Photosynthesis absorbs energy (endothermic). Carbon is reduced and oxygen is oxidised, so it is a redox reaction. <i>Most popular wrong answer: D (27%).</i>	54% correct A 10% · B 54% · C 10% · D 27%
☐	13	A	$\text{Cr}_2\text{O}_7^{2-}$ is reduced, so it is the oxidant. I_2/I^- is a conjugate pair because they differ only by electrons. <i>Most popular wrong answer: B (30%).</i>	49% correct A 49% · B 30% · C 13% · D 8%
☐	14	C	Chemical energy decreases as electrical energy is produced. Electrons flow to the half-cell where Sn^{4+} is reduced at the inert Pt electrode. <i>Most popular wrong answer: D (20%).</i>	54% correct A 14% · B 11% · C 54% · D 20%
☐	15	D	Both statements have a flaw once you check them against how fuel cells work, so neither is correct. <i>Most popular wrong answer: B (40%).</i>	51% correct A 5% · B 40% · C 5% · D 51%

✓	Q	Ans	Why	How Victoria went
☐	16	C	Endothermic means high T favours the products. Choose the pressure from the moles of gas on each side: 100 kPa and 100 °C. <i>Most popular wrong answer: A (15%).</i>	58% correct A 15% · B 11% · C 58% · D 15%
☐	17	C	At t_3 the system has not reached equilibrium and $Q < K$, so the forward rate is greater than the reverse rate. <i>Most popular wrong answer: D (25%).</i>	49% correct A 17% · B 9% · C 49% · D 25%
☐	18	C	Draw every aldehyde isomer (the CHO must be on the end): there are 4, and two of them have a single methyl branch. <i>Most popular wrong answer: A (24%).</i>	51% correct A 24% · B 19% · C 51% · D 7%
☐	19	D	Add the formulas of glycerol and the three fatty acids, then subtract 3 H ₂ O: C ₅₇ H ₈₆ O ₆ . <i>Most popular wrong answer: B (21%).</i>	49% correct A 9% · B 21% · C 21% · D 49%
☐	20	B	Adding water would let hydrolysis compete with the intended reaction and reduce the yield. <i>Most popular wrong answer: C (30%).</i>	59% correct A 8% · B 59% · C 30% · D 3%
☐	21	D	Order by the strength of interaction with the stationary phase: alcohol (H-bonding) > ketone (polar) > alkane. <i>Most popular wrong answer: A (19%).</i>	50% correct A 19% · B 14% · C 17% · D 50%
☐	22	C	A disulfide bridge is a covalent S–S bond between cysteine R groups, so it is part of tertiary structure. <i>Most popular wrong answer: D (19%).</i>	57% correct A 14% · B 11% · C 57% · D 19%
☐	23	A	Photosynthesis is the reverse of respiration: +2840 kJ per 6 mol CO ₂ = 473.3 kJ mol ⁻¹ absorbed. <i>Most popular wrong answer: C (29%).</i>	56% correct A 56% · B 9% · C 29% · D 6%
☐	24	D	Zinc does not react directly with water under these conditions, so the aqueous cell is viable. <i>Most popular wrong answer: C (23%).</i>	57% correct A 11% · B 8% · C 23% · D 57%
☐	25	D	Recharge reverses discharge. The cadmium that was oxidised on discharge is reduced back: $\text{Cd}(\text{OH})_2 + 2e^- \rightarrow \text{Cd} + 2\text{OH}^-$. <i>Most popular wrong answer: A (20%).</i>	61% correct A 20% · B 6% · C 13% · D 61%
☐	26	D	Reversing inverts K. Doubling squares it: $(1/0.286)^2 = 12.2$. <i>Most popular wrong answer: B (21%).</i>	56% correct A 12% · B 21% · C 10% · D 56%
☐	27	C	Both benefits hold, and both improve efficiency. <i>Most popular wrong answer: B (34%).</i>	49% correct A 10% · B 34% · C 49% · D 7%
☐	28	C	Fractional distillation separates liquids with close boiling points. Simple distillation cannot. <i>Most popular wrong answer: D (19%).</i>	59% correct A 14% · B 7% · C 59% · D 19%
☐	29	C	Count the carbons and hydrogens from the skeletal structures: geranial is C ₁₀ H ₁₆ O and linalool is C ₁₀ H ₁₈ O. Each ring or C=O or C=C costs 2 H. <i>Most popular wrong answer: D (24%).</i>	60% correct A 6% · B 10% · C 60% · D 24%
☐	30	B	Each I ₂ adds across one C=C. 6 mol I ₂ per triglyceride means 6 C=C, which is 2 per fatty acid, so it is linoleic acid. <i>Most popular wrong answer: A (32%).</i>	50% correct A 32% · B 50% · C 11% · D 7%

Multiple-choice score: _____ / 30

Section B – Short- and extended-answer questions

Question 1 (7 marks) · state average 3.5 / 7 (50%)

a. Type of reaction converting C₄H₈ to the chloroalkane

☐ Addition

Answer / notes: Addition.

State average 0.6 / 1 (60%); 64% scored full marks. Mark distribution: 0: 36% · 1: 64%

Your mark: _____ / 1

b. Semi-structural formula of the isomer that cannot give a primary alcohol

☐ CH₃CHClCH₂CH₃

Answer / notes: 2-chlorobutane, CH₃CHClCH₂CH₃. 1-chlorobutane is the one that gives the primary alcohol.

State average 0.5 / 1 (50%); 48% scored full marks. Mark distribution: 0: 52% · 1: 48%

Your mark: _____ / 1

c. Reagent to convert the chloroalkane into a primary alcohol

☐ NaOH(aq) or KOH(aq)

Answer / notes: NaOH(aq). Water alone is too slow.

State average 0.4 / 1 (40%); 63% scored full marks. Mark distribution: 0: 37% · 1: 63%

Your mark: _____ / 1

d.i. Catalyst for esterification

☐ Concentrated H₂SO₄ (or H₃PO₄)

Answer / notes: Concentrated H₂SO₄. Writing (aq) lost the mark.

State average 0.4 / 1 (40%); 36% scored full marks. Mark distribution: 0: 64% · 1: 36%

Your mark: _____ / 1

d.ii. Balanced equation for propan-1-ol + methanoic acid

☐ Correct reactants

☐ Correct products (ester + water)

Answer / notes: CH₃CH₂CH₂OH + HCOOH → HCOOCH₂CH₂CH₃ + H₂O

State average 1.0 / 2 (50%); 40% scored full marks. Mark distribution: 0: 36% · 1: 24% · 2: 40%

Your mark: _____ / 2

d.iii. IUPAC name of the ester

☐ Propyl methanoate

Answer / notes: Propyl methanoate.

State average 0.6 / 1 (60%); 61% scored full marks. Mark distribution: 0: 39% · 1: 61%

Your mark: _____ / 1

Question 2 (10 marks) · state average 5.3 / 10 (53%)

a. Pressure in the CO₂ storage tank

☐ Uses PV = nRT with T = 301 K

☐ P = 6.47 × 10³ kPa (3 sig. figs)

Answer / notes: $P = \frac{5.17 \times 10^4 \times 8.31 \times 301}{20000.0} = 6.47 \times 10^3 \text{ kPa}$.

State average 1.3 / 2 (65%); 47% scored full marks. Mark distribution: 0: 19% · 1: 34% · 2: 47%

Your mark: _____ / 2

b.i. Direction of electron flow in the wire, justified

☐ Arrow towards the cathode

☐ Justification: electrons are needed at the cathode for 2H⁺ + 2e⁻ → H₂ (they flow from the anode to the cathode)

Answer / notes: From the anode to the cathode, because the cathode needs electrons to reduce H⁺ to H₂.

State average 1.2 / 2 (60%); 46% scored full marks. Mark distribution: 0: 26% · 1: 28% · 2: 46%

Your mark: _____ / 2

b.ii. Two functions of the membrane

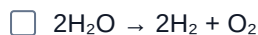
- ☐ Allows H^+ ions to pass through to complete the circuit
- ☐ Keeps the H_2 and O_2 products apart (prevents a spontaneous reaction between them)

Answer / notes: It lets H^+ through and keeps H_2 and O_2 apart. "Separates the reactants" was the common wrong answer.

State average 0.4 / 2 (20%); 6% scored full marks. Mark distribution: 0: 70% · 1: 24% · 2: 6%

Your mark: _____ / 2

c.i. Overall equation for the electrolyser at 80 °C



Answer / notes: $2\text{H}_2\text{O}(l) \rightarrow 2\text{H}_2(g) + \text{O}_2(g)$

State average 0.5 / 1 (50%); 51% scored full marks. Mark distribution: 0: 49% · 1: 51%

Your mark: _____ / 1

c.ii. Moles of H_2 from 1625.0 A for 1.25 h

- ☐ $Q = It = 7.31 \times 10^6 \text{ C}$
- ☐ $n(e^-) = Q/F = 75.8 \text{ mol}$
- ☐ $n(\text{H}_2) = 37.9 \text{ mol}$

Answer / notes: $Q = 1625.0 \times 4500 = 7.31 \times 10^6 \text{ C}$, $n(e^-) = 75.8 \text{ mol}$, and $n(\text{H}_2) = 37.9 \text{ mol}$.

State average 1.9 / 3 (63%); 46% scored full marks. Mark distribution: 0: 19% · 1: 12% · 2: 23% · 3: 46%

Your mark: _____ / 3

Question 3 (10 marks) · state average 4.7 / 10 (47%)

2023 VCAA exam · Section B · Question 7

a.i. Molecular formula of the molecule, justified from the mass spectrum

- ☐ $\text{C}_5\text{H}_{10}\text{O}$
- ☐ Molecular ion at $m/z = 86$, so $M = 86 \text{ g mol}^{-1}$

Answer / notes: $\text{C}_5\text{H}_{10}\text{O}$. The parent ion is at $m/z = 86$.

State average 1.3 / 2 (65%); 60% scored full marks. Mark distribution: 0: 30% · 1: 10% · 2: 60%

Your mark: _____ / 2

a.ii. Why there is a small peak at $m/z = 87$

- ☐ Molecules containing a ^{13}C isotope

Answer / notes: Molecules containing one ^{13}C atom.

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 40% · 1: 70%

Your mark: _____ / 1

b. What the doublet at 1.1 ppm tells you

- ☐ Adjacent carbon has one H ($n + 1$ rule)

Answer / notes: The neighbouring carbon carries a single H, e.g. $(\text{CH}_3)_2\text{CH}-$.

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: _____ / 1

c. Draw a structure consistent with all the data

- ☐ A valid isomer of $\text{C}_5\text{H}_{10}\text{O}$
- ☐ Consistent with 4 C environments / 3 H environments
- ☐ Correct full structural formula of 3-methylbutan-2-one

Answer / notes: 3-methylbutan-2-one: $\text{CH}_3\text{COCH}(\text{CH}_3)_2$, drawn with every bond shown.

State average 1.3 / 3 (43%); 30% scored full marks. Mark distribution: 0: 40% · 1: 30% · 2: 10% · 3: 30%

Your mark: _____ / 3

d. Why one molecule absorbs several IR frequencies

- ☐ Different bonds/functional groups absorb different frequencies
- ☐ Absorption depends on bond strength and the masses of the bonded atoms
- ☐ Different vibration modes (stretching, bending) have different energies

Answer / notes: Each bond type vibrates at its own frequency, set by bond strength and atom mass, and each bond has several vibration modes.

State average 0.9 / 3 (30%); 10% scored full marks. Mark distribution: 0: 30% · 1: 50% · 2: 10% · 3: 10%

Your mark: _____ / 3

Question 4 (8 marks) · state average 3.8 / 8 (48%)

2024 VCAA exam · Section B · Question 1

a. How melting point determination checks the identity and purity of solids

- ☐ A sharp (narrow) melting range means pure, a broad or lowered range means impure
- ☐ Matching the literature melting point of the pure compound (or a mixed-melting-point test) confirms identity

Answer / notes: Pure solids melt sharply at the literature value, and impurities lower and broaden the range.

State average 0.7 / 2 (35%); 11% scored full marks. Mark distribution: 0: 43% · 1: 46% · 2: 11%

Your mark: _____ / 2

b.i. Number of chiral centres in linalool

- ☐ One

Answer / notes: One: the carbon bearing the –OH.

State average 0.7 / 1 (70%); 69% scored full marks. Mark distribution: 0: 31% · 1: 69%

Your mark: _____ / 1

b.ii. % by mass of linalool in a mixture with hexane, from moles of I₂ reacted

- ☐ $n(\text{linalool}) = 0.042/2 = 0.021 \text{ mol}$ (2 C=C per molecule)
- ☐ $m(\text{linalool}) = 0.021 \times 154 = 3.23 \text{ g}$
- ☐ $\% = 3.23/15.0 \times 100 = 22\%$

Answer / notes: Linalool has 2 C=C, so 0.021 mol, which is 3.23 g, which is 22%. Hexane does not react.

State average 1.3 / 3 (43%); 26% scored full marks. Mark distribution: 0: 42% · 1: 13% · 2: 18% · 3: 26%

Your mark: _____ / 3

b.iii. Two processes to obtain purified linalool from lavender

- ☐ An extraction technique (steam distillation, solvent extraction)
- ☐ A purification technique (fractional distillation, HPLC)

Answer / notes: Steam distillation to extract it, then fractional distillation to purify it. Recrystallisation doesn't suit an oil.

State average 1.1 / 2 (55%); 29% scored full marks. Mark distribution: 0: 18% · 1: 54% · 2: 29%

Your mark: _____ / 2

Question 5 (8 marks) · state average 3.7 / 8 (46%)

2024 VCAA exam · Section B · Question 4

a. Expression for K_p

☐ $K_p = \frac{p_{\text{NH}_3}^2}{p_{\text{N}_2} p_{\text{H}_2}^3}$

Answer / notes: $K_p = \frac{p_{\text{NH}_3}^2}{p_{\text{N}_2} p_{\text{H}_2}^3}$, the same shape as K_c but using partial pressures.

State average 0.9 / 1 (90%); 86% scored full marks. Mark distribution: 0: 14% · 1: 86%

Your mark: _____ / 1

b. Partial pressure of NH₃ at equilibrium

- ☐ 0.696 kPa

Answer / notes: $p_{\text{NH}_3} = \sqrt{1.60 \times 10^{-8} \times 43.4 \times 88.7^3} = 0.696 \text{ kPa}$.

State average 0.4 / 1 (40%); 34% scored full marks. Mark distribution: 0: 66% · 1: 34%

Your mark: _____ / 1

c. A change that makes $Q = 1.20 \times 10^{-9}$ and how the system responds

- ☐ A valid change making $Q < K$ (remove NH_3 , add N_2/H_2 , or decrease the volume)
- ☐ The equilibrium shifts forward (right)
- ☐ Until Q rises back to K / partially restores the change

Answer / notes: $Q < K$, so something like removing NH_3 happened, and the system shifts right until $Q = K$.

State average 1.1 / 3 (37%); 13% scored full marks. Mark distribution: 0: 33% · 1: 38% · 2: 16% · 3: 13%

Your mark: _____ / 3

d. How the iron catalyst resolves the rate–temperature conflict, with a green principle

- ☐ The conflict: high T gives a fast rate but a low yield (exothermic), while low T gives a high yield but a slow rate
- ☐ The catalyst lowers E_a , so an acceptable rate is reached at a lower temperature (better yield)
- ☐ Links to a green principle (catalysis, or design for energy efficiency)

Answer / notes: The catalyst speeds up the reaction at a moderate temperature, so the yield isn't sacrificed. Link it to the "catalysis" or "energy efficiency" principle.

State average 1.3 / 3 (43%); 18% scored full marks. Mark distribution: 0: 23% · 1: 41% · 2: 18% · 3: 18%

Your mark: _____ / 3

Question 6 (11 marks) · state average 5.5 / 11 (50%)

2024 VCAA exam · Section B · Question 5

a.i. Energy released by burning 0.580 g of butane

- ☐ $n(\text{C}_4\text{H}_{10}) = 0.0100 \text{ mol}$
- ☐ Energy = 28.8 kJ with units

Answer / notes: $0.580/58.0 = 0.0100 \text{ mol}$, and $\times 2880 = 28.8 \text{ kJ}$.

State average 1.5 / 2 (75%); 62% scored full marks. Mark distribution: 0: 19% · 1: 20% · 2: 62%

Your mark: _____ / 2

a.ii. Energy absorbed by the 100.0 g of water

- ☐ 14.9 kJ with units

Answer / notes: $100.0 \times 4.18 \times 35.6 = 14.9 \text{ kJ}$. Don't use the density of water here: the mass is given.

State average 0.7 / 1 (70%); 66% scored full marks. Mark distribution: 0: 34% · 1: 66%

Your mark: _____ / 1

a.iii. Percentage energy efficiency

- ☐ Correct calculation ($14.9/28.8 \times 100$)
- ☐ Quoted to 3 sig. figs: 51.7%

Answer / notes: 51.7%.

State average 1.2 / 2 (60%); 51% scored full marks. Mark distribution: 0: 29% · 1: 20% · 2: 51%

Your mark: _____ / 2

b.i. Why a calorimeter must be calibrated

- ☐ Some energy heats the calorimeter itself (container, thermometer, stirrer), not just the water, so $q = mc\Delta T$ is not enough
- ☐ The calibration factor ($\text{J } ^\circ\text{C}^{-1}$) links the observed ΔT to the energy actually released

Answer / notes: Energy goes into the whole calorimeter, not just the water, so you need the calibration factor to turn ΔT into energy.

State average 0.6 / 2 (30%); 9% scored full marks. Mark distribution: 0: 56% · 1: 35% · 2: 9%

Your mark: _____ / 2

b.ii. Maximum temperature if 0.048 mol of KI reacts

- ☐ Energy released = $0.048/2 \times 49.0 = 1.18 \text{ kJ}$ (uses the 2:1 ratio)
- ☐ $\Delta T = 1176/470.0 = 2.5 \text{ }^{\circ}\text{C}$
- ☐ Maximum temperature = $27.5 \text{ }^{\circ}\text{C}$

Answer / notes: ΔH is per 2 mol of KI, so $0.024 \times 49.0 = 1.18 \text{ kJ}$. $\Delta T = 2.5 \text{ }^{\circ}\text{C}$, which gives $27.5 \text{ }^{\circ}\text{C}$. Ignoring the stoichiometry was the common error.

State average 1.3 / 3 (43%); 23% scored full marks. Mark distribution: 0: 43% · 1: 9% · 2: 24% · 3: 23%

Your mark: _____ / 3

b.iii. A reason the average maximum temperature was higher than expected

- ☐ A valid reason (e.g. the KI concentration was higher than stated, more KI was added, a smaller total volume than during calibration, or the calibration factor was too small)

Answer / notes: For example, the KI stock was more concentrated than 0.960 M, so more reacted. 87% scored zero because heat loss makes the result lower, not higher.

State average 0.2 / 1 (20%); 13% scored full marks. Mark distribution: 0: 87% · 1: 13%

Your mark: _____ / 1

Question 7 (13 marks) · state average 6.4 / 13 (49%)

2024 VCAA exam · Section B · Question 6

a. Bond responsible for the highest-wavenumber IR absorption

- ☐ O–H (alcohol)

Answer / notes: O–H (alcohol), the broad band near 3300 cm^{-1} .

State average 0.7 / 1 (70%); 68% scored full marks. Mark distribution: 0: 32% · 1: 68%

Your mark: _____ / 1

b.i. m/z of the base peak

- ☐ 56

Answer / notes: 56. Don't write m/z as a unit.

State average 0.8 / 1 (80%); 75% scored full marks. Mark distribution: 0: 25% · 1: 75%

Your mark: _____ / 1

b.ii. Formula of the species at $m/z = 73$

- ☐ $[\text{C}_4\text{H}_9\text{O}]^+$ (positive charge shown)

Answer / notes: $[\text{C}_4\text{H}_9\text{O}]^+$. Show the + charge.

State average 0.5 / 1 (50%); 50% scored full marks. Mark distribution: 0: 50% · 1: 50%

Your mark: _____ / 1

c. What four ^{13}C peaks say about five carbons

- ☐ Four carbon environments
- ☐ So two carbons are in the same environment (equivalent)
- ☐ So the molecule is branched / symmetric (or a peak is matched to a carbon type)

Answer / notes: There are 4 environments for 5 carbons, so two are equivalent, which means a symmetric branch such as two identical CH_3 groups.

State average 1.7 / 3 (57%); 19% scored full marks. Mark distribution: 0: 14% · 1: 26% · 2: 41% · 3: 19%

Your mark: _____ / 3

d.i. What an integration curve represents

- ☐ The area under the peak (signal) for a hydrogen environment

Answer / notes: The area of that peak. 80% confused this with part ii.

State average 0.2 / 1 (20%); 20% scored full marks. Mark distribution: 0: 80% · 1: 20%

Your mark: _____ / 1

d.ii. What the relative sizes of the integration curves tell you

- ☐ The ratio of the numbers of hydrogens in each environment

Answer / notes: The relative number of H atoms in each environment.

State average 0.3 / 1 (30%); 32% scored full marks. Mark distribution: 0: 68% · 1: 32%

Your mark: _____ / 1

d.iii. What the given integration values indicate

- ☐ The H ratio across the four environments is 1:2:2:6

Answer / notes: 1:2:2:6 hydrogens across the four environments.

State average 0.2 / 1 (20%); 20% scored full marks. Mark distribution: 0: 80% · 1: 20%

Your mark: _____ / 1

d.iv. What the 0.96 ppm peak (shift and size) indicates

- ☐ R-CH₃ environment
☐ Six H there, so two equivalent CH₃ groups

Answer / notes: Two equivalent CH₃ groups on a carbon with no H (hence a singlet).

State average 1.1 / 2 (55%); 19% scored full marks. Mark distribution: 0: 13% · 1: 68% · 2: 19%

Your mark: _____ / 2

e. Draw Compound X

- ☐ Any alcohol isomer of C₅H₁₁ClO
☐ Correct structure: 3-chloro-2,2-dimethylpropan-1-ol

Answer / notes: A central C bearing two CH₃ groups, one CH₂Cl and one CH₂OH.

State average 0.9 / 2 (45%); 20% scored full marks. Mark distribution: 0: 31% · 1: 49% · 2: 20%

Your mark: _____ / 2

Question 8 (12 marks) · state average 6.4 / 12 (53%)

2024 VCAA exam · Section B · Question 7

a.i. Which electrode is the anode

- ☐ Fe

Answer / notes: Fe (connected to the positive terminal).

State average 0.8 / 1 (80%); 76% scored full marks. Mark distribution: 0: 24% · 1: 76%

Your mark: _____ / 1

a.ii. Half-equation at the Pt electrode

- ☐ $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$ (with states, single arrow)

Answer / notes: $2\text{H}_2\text{O}(\text{l}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$. Water is the strongest oxidant present, since Na⁺ and NO₃⁻ are weaker.

State average 0.3 / 1 (30%); 25% scored full marks. Mark distribution: 0: 75% · 1: 25%

Your mark: _____ / 1

a.iii. Why no reaction occurs once the power is off

- ☐ No oxidant–reductant pair present can react spontaneously (e.g. the H₂ product has escaped, and Fe with the remaining species gives no spontaneous reaction)

Answer / notes: The products don't stay in contact (H₂ bubbles off), and nothing left can react spontaneously. 94% scored zero.

State average 0.1 / 1 (10%); 6% scored full marks. Mark distribution: 0: 94% · 1: 6%

Your mark: _____ / 1

b.i. Polarity of electrode X during recharge

- ☐ Positive

Answer / notes: Positive. Oxidation happens there, so it is the anode in an electrolytic cell.

State average 0.7 / 1 (70%); 69% scored full marks. Mark distribution: 0: 31% · 1: 69%

Your mark: _____ / 1

b.ii. Mass of LiC_6 produced at 1.25 A for 1.00 h

- ☐ $Q = 4500 \text{ C}$
- ☐ $n(\text{e}^-) = 0.0466 \text{ mol}$
- ☐ $n(\text{LiC}_6) = 0.0466 \text{ mol (1:1)}$
- ☐ $m = 0.0466 \times 78.9 = 3.68 \text{ g}$

Answer / notes: $4500/96\,500 = 0.0466 \text{ mol}$, which gives 3.68 g.

State average 3.1 / 4 (78%); 60% scored full marks. Mark distribution: 0: 15% · 1: 3% · 2: 4% · 3: 18% · 4: 60%

Your mark: _____ / 4

c.i. Overall energy transformation in artificial photosynthesis

- ☐ Light (solar) energy to chemical energy

Answer / notes: Light energy to chemical energy.

State average 0.4 / 1 (40%); 33% scored full marks. Mark distribution: 0: 67% · 1: 33%

Your mark: _____ / 1

c.ii. Direction of electron flow on the diagram

- ☐ Arrow from the anode (water oxidation) to the cathode (H^+ reduction)

Answer / notes: From where water is oxidised to where H^+ is reduced.

State average 0.6 / 1 (60%); 59% scored full marks. Mark distribution: 0: 41% · 1: 59%

Your mark: _____ / 1

c.iii. How charge stays balanced when electrons flow one way

- ☐ H^+ ions produced at the anode migrate to the cathode
- ☐ where they are consumed, balancing the electrons arriving through the circuit

Answer / notes: H^+ made at the anode travels through the electrolyte to the cathode, matching the electrons that go through the wire.

State average 0.4 / 2 (20%); 7% scored full marks. Mark distribution: 0: 68% · 1: 25% · 2: 7%

Your mark: _____ / 2

Question 9 (8 marks) · state average 4.3 / 8 (54%)

2025 VCAA exam · Section B · Question 7

a. A requirement for molten oxide electrolysis to help UN SDG 12

- ☐ Identifies a key issue (e.g. the huge electricity demand, or rare anode materials)
- ☐ Links it to a requirement (the electricity must come from renewables, or anode materials must be sustainably sourced)

Answer / notes: MOE uses about $17\times$ more electricity, so it only helps if that electricity is renewable.

State average 0.7 / 2 (35%); 15% scored full marks. Mark distribution: 0: 53% · 1: 32% · 2: 15%

Your mark: _____ / 2

b. Evaluate both processes against two green principles, include an ethical factor, and conclude

- ☐ Identifies a relevant green chemistry principle
- ☐ Compares both methods against it using the data
- ☐ Identifies a second relevant principle
- ☐ Compares both methods against it using the data
- ☐ A relevant ethical factor (e.g. jobs/communities, access to steel, climate impact on future generations)
- ☐ A conclusion consistent with the evaluation

Answer / notes: For example, atom economy (MOE produces only O_2 as a by-product) and energy efficiency (the blast furnace uses less electricity but emits CO_2). Add an ethical point and a clear conclusion.

State average 3.6 / 6 (60%); 15% scored full marks. Mark distribution: 0: 11% · 1: 5% · 2: 11% · 3: 15% · 4: 20% · 5: 22% · 6: 15%

Your mark: _____ / 6

Your result

Total: _____ / 117 State average on these questions: **60 / 117 (51%)**

Topics to revise:

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